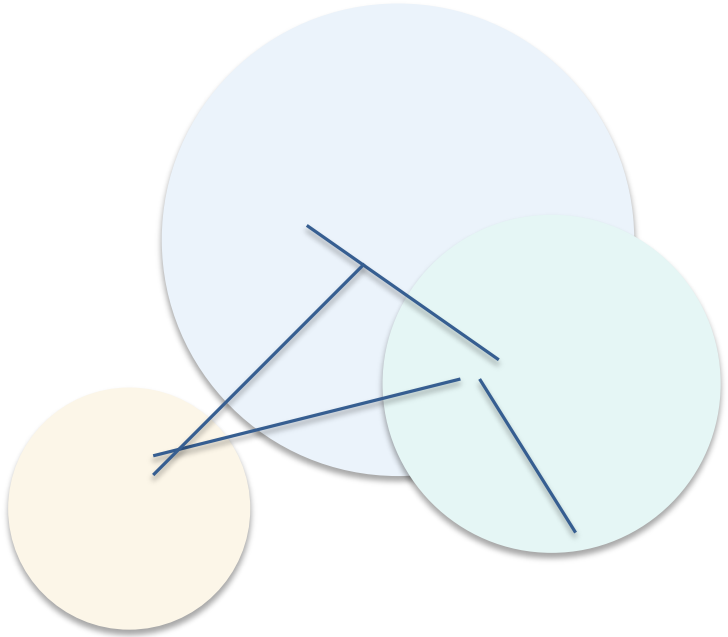
RESEARCH PAPER REVIEW

Specialized Decision Models vs. General-Purpose LLMs

Benchmarking Jev Across Knowledge, Reasoning, and Multilingual Tasks



Presented by

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A comparative look at specialized decisions and general-purpose language models

The Problem: Do All AI Decisions Need a General-Purpose LLM?

WHY THIS STUDY?

The researchers explore whether every structured decision requires a broad, general-purpose language model.

General-purpose LLMs

- Can handle many different tasks.
- Are increasingly used for text classification and annotation.
- May be more than what is needed for some simple, specific decisions.

Specialized models

- Are designed or adapted for particular tasks.
- Can perform better on certain tasks, even when smaller.
- May be suitable alternatives for specific decisions.

RESEARCH QUESTION

Can a specialized decision model like Jev handle certain decisions effectively without needing a general-purpose LLM every time?

Meet Jev: A System 1 Decision Model

SPECIALIZED DECISION MODEL

A specialized System 1 decision model

What is Jev?

- **Jev is a specialized decision model designed to answer structured questions about an input.**
- **It follows a System 1 approach — making direct, intuitive decisions rather than relying on deliberate, step-by-step reasoning.**
- **It is designed for specific decision-making tasks rather than general-purpose text generation.**

Key features

Choice

Selects one option from up to 255 candidates.

Score

Rates an input using an ordered scoring scale.

Noul

Estimates the probability of a yes/no statement.

Research Objective and Key Questions

WHAT THE STUDY INVESTIGATES

Main objective

To evaluate how Jev performs compared with 19 general-purpose LLMs across knowledge, reasoning, and multilingual tasks.

1

Knowledge

Can Jev perform as well as general-purpose LLMs on knowledge-based questions?

2

Reasoning

How does Jev perform on tasks requiring mathematical calculations and multi-step reasoning?

3

Multilingual ability

How well does Jev handle questions across different languages?

4

Practical use

Which types of decisions can Jev handle effectively on its own?

Overall goal: Identify Jev's strengths and limitations to understand where a specialized decision model may be useful.

Understanding the 13 Benchmarks

HOW JEV IS TESTED

A benchmark is a standardized set of questions used to evaluate and compare models.

8 BENCHMARKS

Knowledge

Tests academic knowledge, commonsense, and understanding of information.

Examples: MMLU, MMLU Redux , MMLU Pro , GPQA Diamond, C-Eval , WindoGrande, TruthfulQA

3 BENCHMARKS

Reasoning

Tests problem-solving, mathematical calculations, and multi-step reasoning.

Examples: ARC-Challenge, MathQA, AQuA-RAT.

2 BENCHMARKS

Multilingual

Tests performance across multiple languages.

Examples: MMLU-ProX (29 languages) and MMMLU (14 languages).

Why these benchmarks? They allow researchers to compare Jev and general-purpose LLMs across different capabilities using standardized question sets.

The 19 LLMs: Three Tiers of Comparison

COMPARISON GROUPS

The models are grouped into three tiers for comparison with Jev.

6 MODELS

Frontier LLMs

Recently released models.

GPT-5.6 Sol
Claude Opus 5.5
Gemini 3.8 Flash
Qwen 3.8 Flash
Qwen3.8-Max
DeepSeek-V4.1-Flash

7 MODELS

Representative LLMs

Widely used earlier models.

GPT-4o
Claude 3.5 Sonnet
DeepSeek-R1
DeepSeek – V3
Qwen3 32B
Qwen 2.5 – 72 B Instruct
Llama-3.1-405B-Instruct

6 MODELS

Small LLMs

Models with fewer than 10 billion parameters.

Qwen3-8B
Qwen 3 – 4B
Llama-3.1-8B-Instruct
Phi-4-mini-instruct
Gemma-3-4B-IT
Mistral-7B-v0.3

Why compare three tiers? To examine Jev's performance against newer models, established models, and smaller models.

How the Evaluation Works

FROM QUESTIONS TO ACCURACY

The researchers evaluate each model using standardized multiple-choice questions and compare predictions with the answer key.

1

Benchmark Questions

Use standardized multiple-choice questions with correct answers.

2

Model Prediction

Jev uses Choice to select an option; LLMs answer the same benchmark questions.

3

Compare with Answer Key

Check whether each selected answer is correct.

4

Calculate Accuracy

Correct answers divided by total questions, multiplied by 100%.

Accuracy = (Correct answers ÷ Total questions) × 100%

Example: 85 correct answers out of 100 = 85% accuracy.

Important details from the paper

- **Each Jev and frontier LLM evaluation was run once.**
- **Of 169 representative/small model–benchmark pairs, 104 used published scores and 65 were evaluated by the researchers.**
- **Published results may use different prompts and settings, so some comparisons are approximate.**

Overall Results: How Does Jev Compare?

MAIN FINDINGS

The overall results show a mix of clear strengths and task-specific limitations.

Strong performance

- Achieved the best score among all evaluated models on MMLU-Redux and ARC-Challenge.
- Performed competitively on several knowledge and commonsense benchmarks.

Weakness in mathematics

- Fell behind frontier LLMs on MathQA and AQuA-RAT.
- Scored below all 19 general-purpose LLMs on MathQA.

Multilingual performance

- Performed well on translated multilingual questions.
- Had more difficulty with C-Eval, which contains questions from Chinese exams.

Main takeaway **Jev performs well on many knowledge-based tasks, but mathematical problem-solving remains a major limitation.**

Knowledge vs. Mathematics: Where Jev Excels and Struggles

A KEY PERFORMANCE CONTRAST

Knowledge-based tasks

- Strong performance on several knowledge and commonsense benchmarks.
- Outperformed all small LLMs on the non-math comparisons.
- Achieved the best score on MMLU-Redux.

Mathematical tasks

- Struggled with multi-step math word problems.
- Fell below all 19 general-purpose LLMs on MathQA.
- Scored 17.7 percentage points below the frontier median on MathQA.

Jev on MathQA	75.0%
Frontier-model median	92.7%

Gap: −17.7 percentage points

Why might this happen?

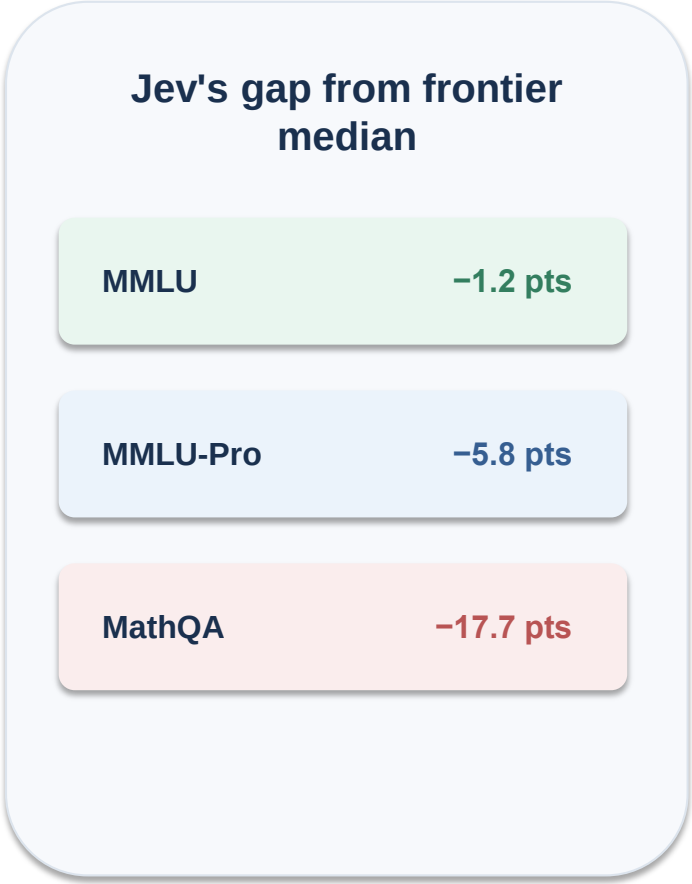
Knowledge questions may involve identifying an option, while math word problems can require several calculation steps. The authors suggest multi-step calculation may be a weakness for Jev.

Reasoning Demands: How Does Task Complexity Affect Jev?

PERFORMANCE DIFFERENCES ACROSS TASKS

Key findings

- **The reasoning gap grows:** Jev's gap from the frontier-model median is small on MMLU, moderate on MMLU-Pro, and large on GPQA Diamond, AQuA-RAT, and MathQA.
- **MMLU vs. MMLU-Pro:** Jev is 1.2 percentage points below the frontier median on MMLU, compared with 5.8 points below on MMLU-Pro.
- **Chain-of-thought prompting:** In a cited study, it improved GPT-4o's score by 1.5 points on MMLU and 19.1 points on MMLU-Pro.
- **DeepSeek-R1:** The strongest representative LLM in the paper, slightly ahead of Jev on average. It also leads Jev on MMLU-Pro, C-Eval, HellaSwag, and MMMLU, apart from the two math benchmarks.



Main takeaway **Jev's performance gap tends to increase on tasks requiring more calculation and multi-step reasoning.**

Multilingual Performance: Translated vs. Native Questions

LANGUAGE AND QUESTION ORIGIN

The study distinguishes translated benchmark questions from questions drawn from Chinese exams.

Translated questions

- MMMLU: MMLU translated into 14 languages.
- MMLU-ProX: MMLU-Pro translated into 29 languages.
- Jev performs well on translated multilingual questions, remaining competitive with frontier LLMs on MMLU-ProX.

Questions from Chinese exams

- C-Eval: Covers 52 disciplines in Chinese.
- Jev scores about 8 percentage points below the frontier median.
- It also falls behind all representative LLMs on C-Eval.

Main takeaway **Jev transfers reasonably well to translated questions, but struggles more with knowledge specific to Chinese exams.**

Reliability Check: Comparing Independent Evaluations

CROSS-EVALUATION COMPARISON

The researchers compared their results with an independent study by Deußer et al. (2026) across five shared benchmarks.

Benchmark	This paper	Independent evaluation	Difference
MMLU	91.3%	91.8%	0.5 pts
C-Eval	85.3%	83.9%	1.4 pts
HellaSwag	89.3%	95.5%	6.2 pts
WinoGrande	91.8%	91.4%	0.4 pts
ARC	97.6%	98.8%	1.2 pts*

Why did HellaSwag differ?

The authors suggest that different request templates may have affected Jev's results, because it is sensitive to how answer options are presented.

How to interpret ARC

The ARC comparison is not like-for-like: the independent result combines ARC-Easy and ARC-Challenge.

Main takeaway: Jev's results were broadly consistent across evaluations, but question and option presentation can influence performance.

Future Directions: Making Jev More Useful

WHAT COULD COME NEXT?

1. Evaluate all models under one protocol

- Test all model tiers using the same evaluation approach.
- Make comparisons more consistent and reliable.

2. Use Jev's probabilities to route questions

- Investigate whether Jev's option probabilities can identify questions that need deeper reasoning.
- Pass such questions to a reasoning-focused model when needed.

Proposed approach

Jev handles suitable decisions → **Assess whether deeper reasoning is needed** → **Reasoning model handles complex questions**

This is a proposed future approach, not a system demonstrated by the paper.

Conclusion: The Right Model for the Right Task

KEY TAKEAWAYS

Knowledge

Jev performs well on several knowledge and commonsense benchmarks.

Mathematics

It struggles compared with general-purpose LLMs on mathematical problem-solving.

Multilingual tasks

It performs reasonably well on translated questions but faces challenges with Chinese exam-specific knowledge.

Practical use

Jev may be useful for specific decision-making tasks, while complex questions may benefit from models with stronger reasoning capabilities.

The best model depends on the task, not just on whether it is a specialized model or a general-purpose LLM.

Thank you